

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAMINATION
August, 2017

Probability (Ph.D. Version)

Instructions to the Student

1. Answer all six questions. Each will be graded from 0 to 10.
 2. Use a different booklet for each question. Write the problem number and your code number (**NOT YOUR NAME**) on the outside cover.
 3. Keep scratch work on separate pages in the same booklet.
 4. If you use a “well known” theorem in your solution to any problem, it is your responsibility to make clear which theorem you are using and to justify its use.
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1. Let p_n be the probability that in a sequence of independent trials with probability of success $p \in (0, 1)$ there is an odd number of successes in n trials. Determine a formula for p_n , for every $n \geq 1$.

2. Let $\{\xi_n\}_{n \in \mathbb{N}}$ be a sequence of independent exponential random variables with parameter $\lambda = 2$. Find a numeric sequence $\{\alpha_n\}_{n \in \mathbb{N}}$ such that

$$\lim_{n \rightarrow \infty} \frac{\max(\xi_1, \dots, \xi_n)}{\alpha_n} = 1, \quad \text{in probability.}$$

3. Let $\{\xi_n\}_{n \in \mathbb{N}}$ be a sequence of independent random variables with distribution functions $\{F_n\}_{n \in \mathbb{N}}$. Prove that

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \xi_n = 0\right) = 1 \iff \sum_{n=1}^{\infty} (1 - F_n(\epsilon) + F_n(-\epsilon)) < \infty, \quad \forall \epsilon > 0.$$

4. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\{\xi_n\}_{n \in \mathbb{N}}$ be a sequence of random variables in $L^2(\Omega)$. Prove or disprove the following statements.

1. If $\{\xi_n\}_{n \in \mathbb{N}}$ converges in $L^2(\Omega)$, then it converges in probability.

2. If $\{\xi_n\}_{n \in \mathbb{N}}$ is bounded in $L^2(\Omega)$ and converges in probability, then it converges in $L^1(\Omega)$.
3. If $\{\xi_n\}_{n \in \mathbb{N}}$ is bounded in $L^2(\Omega)$ and converges in probability, then it converges in $L^2(\Omega)$.

5. Let $\xi_1, \xi_2, \dots$ be independent identically distributed random variables with expectation 0 and variance 1. Define

$$X_n = \sum_{1 \leq i < j \leq n} \xi_i \xi_j.$$

Find numerical sequences a_n and b_n such that $(X_n - a_n)/b_n$ converges to a non-degenerate distribution and identify the limiting distribution.

6. Let B_t be a Brownian motion, on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with respect to the filtration $\{\mathcal{F}_t\}_{t \geq 0}$.

1. Prove that for every $\theta \in \mathbb{R}$

$$X(t) = \exp \left(\theta B_t - \frac{1}{2} \theta^2 t \right), \quad t \geq 0,$$

is a martingale.

2. Let $a < 0$ and $b \geq 0$ two given constants and let

$$\tau := \inf \{ t \geq 0 : B_t = at + b \}.$$

Prove that $\mathbb{P}(\tau < \infty) = 1$.

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Probability (Masters Version)

Instructions to the Student

- a. Answer all six questions. Each will be graded from 0 to 10.
 - b. Use a different booklet for each question. Write the problem number and your code number (**NOT YOUR NAME**) on the outside cover.
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 - d. If you use a “well known” theorem in your solution to any problem, it is your responsibility to make clear which theorem you are using and to justify its use.
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1. Let X and Y be two independent random variables, exponentially distributed of parameter $\lambda > 0$. Determine the distribution function of

$$U := \frac{X}{\max(X, Y)}.$$

2. Two players, A and B, each have their own box urn, containing white and red balls. Let p_A and p_B be the proportion of white balls for the players A and B, respectively.

The two players, take turns in extracting balls from their urn with replacement, by following the following rules: 1. The player A starts, 2. The winner is the first player who gets a white ball.

1. Compute the probability of winning for each of the players.
2. How do p_A and p_B need to be chosen in order that the game is fair?

3. Let X be a random variable having symmetric distribution (that is $-X$ is distributed as X), such that $\mathbf{E}(X^2) < \infty$, and let Z be a Bernoulli random variable with parameter p , independent of X . We define

$$Y = \begin{cases} X, & \text{if } Z = 1, \\ -X & \text{if } Z = 0. \end{cases}$$

1. Prove that Y and Z are independent and Y has the same distribution as X .
2. For which values of p are X and Y uncorrelated?
3. For which values of p are X and Y independent?

4. A biased coin is tossed n times and the probability that, at each toss, the coin ends up on head is p . A urn contains as many white balls and black balls, as the number of heads and tails, respectively, obtained from the n tosses of the coin. Two balls are extracted from the urn without replacement and they are both white. Given this information, compute the probability that the urn contained r white balls.

5. Let p_n be the probability that in a sequence of independent trials with probability of success $p \in (0, 1)$ there is an odd number of successes in n trials. Prove that

$$p_n = \frac{1}{2} - \frac{1}{2}(1 - 2p)^n, \quad n \geq 1.$$

6. Let $\{\xi_n\}_{n \in \mathbb{N}}$ be a sequence of independent exponential random variables of parameter $\lambda = 1$. Prove that

$$\lim_{n \rightarrow \infty} \frac{\max(\xi_1, \dots, \xi_n)}{\ln n} = 1, \quad \text{in probability.}$$